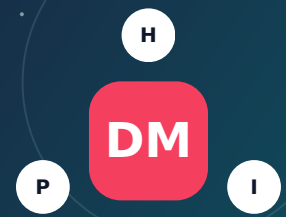


WATER AND INFRASTRUCTURE HAZARD

Dams & Inundation

An inundation map shows where water could travel under a modeled dam emergency; it does not show the annual probability of failure.



1 Understand the factor

What it is - and why it matters

Dam-inundation areas are modeled downstream zones that could be flooded after a dam breach, spillway failure, operational emergency, or unusually large release. Maps often assume a specific reservoir level, breach shape, weather condition, and downstream channel. Water depth, velocity, arrival time, debris, and evacuation access can vary dramatically. Most dams do not fail, but consequences can be severe because warning time may be short and downstream roads, bridges, utilities, and communications can fail together.

How risk can reach a household

- Rapid deep flooding can sweep away structures, vehicles, people, soil, and large debris.
- Roads and bridges may be cut before water reaches a house, limiting evacuation and emergency response.
- Reservoir sediment, sewage, fuel, chemicals, and debris can contaminate the flood zone.
- Emergency releases that do not involve a dam collapse can still cause downstream flooding and prolonged evacuation.

What people often miss

- An inundation boundary is scenario-based and should not be interpreted as a frequent-flood zone or precise edge.
- A distant home can be at greater risk than a closer elevated home because terrain controls flow paths.
- More than one dam, levee, reservoir, or canal may affect the same community.
- Evacuation-route capacity and warning systems may be more important to life safety than structural protection at the house.

? Five checks before buying

1

Identify every upstream dam and obtain official inundation, emergency-action, evacuation, and arrival-time information.

2

Determine the home's elevation, expected depth and velocity, and whether roads, bridges, schools, or hospitals lie in the zone.

3

Review local alert systems, sirens, evacuation routes, assembly points, and plans for people without vehicles.

4

Ask for flood history and inspect the home for prior water damage, raised systems, foundation scour, and sediment lines.

5

Obtain flood-insurance quotes because damage caused by released water is usually treated as flood, regardless of the source.

IMPORTANT DISTINCTION

Map proximity is a screening signal - not proof of exposure, damage, or insurability. Confirm the exact feature, current condition, pathway, and property-specific controls.

HEALTH, PROPERTY AND COVERAGE

How this factor can affect a household

FACTOR 05

H Health impacts

- Life-safety risks include drowning, impact from debris, falls, electrocution, and delayed rescue.
- Evacuation and uncertainty can disrupt medications, caregiving, schooling, work, and mental health.
- Floodwater can contain sewage, fuel, industrial chemicals, pathogens, sharp objects, and contaminated sediment.
- Post-flood dampness, mold, dust, and damaged sanitation systems can extend health risks through recovery.

P Property impacts

- High-velocity water can cause total structural loss, foundation scour, erosion, utility damage, and land loss.
- Downstream infrastructure failure can leave undamaged homes without access, water, sewer, power, or communications.
- Even shallow water can destroy building systems, contents, flooring, insulation, cabinets, and electrical equipment.
- An inundation designation can affect disclosure, emergency planning, buyer perception, financing, and flood-insurance decisions.

I Insurance implications

- Standard homeowners insurance generally excludes flood, including water released by a dam or reservoir.
- Evacuation expenses or loss of use without direct covered damage may not be reimbursed.
- Flood policies may have limits for contents, basements, temporary housing, landscaping, and earth movement or erosion.
- Owners should confirm whether private flood policies address replacement cost, debris removal, code upgrades, and additional living expense.

Possible long-term health implications of buying nearby

- The long-term health concern is event-driven: drowning risk, contaminated water, displacement, mold, and traumatic stress after an emergency.
- Infrastructure outages can be especially dangerous for people who need powered medical devices, refrigeration, dialysis, or home care.
- Residents may experience persistent anxiety when spillways are damaged, reservoirs are high, or warnings recur during storms.
- Understanding warning time and evacuation routes is a core health precaution for every household in a modeled inundation area.

Evidence lens

Dam-failure consequences are well established, while the likelihood of a particular scenario is engineering- and site-specific. Use official maps and emergency plans rather than estimating from distance alone.

Plan more carefully for

Children

Older adults

Pregnancy

Heart/lung conditions

Mobility or power needs

Insurance reality: Availability, eligibility, exclusions, deductibles, and limits vary by state, carrier, property, and cause of loss. Get an address-specific written quote before the purchase contingency expires.

PRACTICAL RISK REDUCTION

Precautions and a real-life example

FACTOR 05

V Verify the actual risk

- Check the National Inventory of Dams and state or local emergency-management maps for all upstream structures.
- Confirm map scenario, reservoir assumptions, expected depth, arrival time, evacuation zone, and latest revision date.

P Protect the property

- Elevate critical systems and documents where practical; use flood-resistant materials in lower areas.
- Maintain drainage and avoid storing fuels, chemicals, or irreplaceable items in likely flow paths.

M Monitor and respond

- Enroll in local alerts, know multiple uphill routes, and evacuate immediately when officials direct it.
- Keep go-bags, medications, chargers, pet supplies, paper maps, and family communication plans ready.

I Prepare insurance records

- Purchase adequate flood coverage and review waiting periods before storm season or reservoir emergencies.
- Document contents and understand limits for basements, temporary housing, debris, and code compliance.

REAL-LIFE EXAMPLE

Oroville Dam Spillway Emergency

February 2017

Damage to the main spillway at California's Oroville Dam and concern about erosion near the emergency spillway led officials to order the temporary evacuation of roughly 188,000 downstream residents. The dam itself did not collapse, but the incident disrupted communities and required major emergency and reconstruction work.

Why it matters: The emergency showed that spillway and operational failures can create mass evacuation and downstream consequences even without a full dam breach.

[Open official incident record - water.ca.gov >](https://water.ca.gov)

Questions to resolve before making a decision

1. Identify every upstream dam and obtain official inundation, emergency-action, evacuation, and arrival-time information.
2. Determine the home's elevation, expected depth and velocity, and whether roads, bridges, schools, or hospitals lie in the zone.
3. Review local alert systems, sirens, evacuation routes, assembly points, and plans for people without vehicles.
4. Ask for flood history and inspect the home for prior water damage, raised systems, foundation scour, and sediment lines.

Official sources and mapping tools

[National Inventory of Dams - nid.sec.usace.army.mil >](https://nid.sec.usace.army.mil)[California DWR Oroville forensic review - water.ca.gov >](https://water.ca.gov)[Ready.gov flooding guidance - ready.gov >](https://ready.gov)